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← Finding shortest paths in weighted graphs

(/learn/advanced-data-structures/module/AfvLN/discussions?sort=lastActivityAtDesc&page=1&q=week3%20quiz)

DJ Possible error in end of quiz after programming exercise.

Djiofack Innocent Finding shortest paths in weighted graphs (/learn/advanced-data-structures/module/AfvLN/discussions) · 10 days ago
 (/learn/advanced-data-structures/module/AfvLN/discussions/v5S9IarVEeWF2Q53QdZUbw)

Hi,
 structures/profiles/a3da4d1cc426cbc43f595fde800768f0)

There is a possible error in the first question about the number of steps taken by djistra and A*. In fact, the end point is exactly the same as the start point.

```
GeographicPoint start = new GeographicPoint(32.868629, -117.215393);
GeographicPoint end = new GeographicPoint(32.868629, -117.215393);
```

The way I have mad my algorithm, this will not even go through the child since we are give the exact same point as begining and start.

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Earliest

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 (/learn/advanced-data-structures/profiles/a3da4d1cc426cbc43f595fde800768f0)
 yep I figured this too , the number of "visited" nodes should be one less than the number of edges visited.

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Cheng Chen · 9 days ago (/learn/advanced-data-structures/module/AfvLN/discussions/v5S9IarVEeWF2Q53QdZUbw/replies/GQfEkaw9EeW2dA51rBQ9EW)
 (/learn/advanced-data-structures/profiles/31aff96e4957af37c7f5213aea266308)
 Actually in the question, they write:

```
GeographicPoint start = new GeographicPoint(32.8648772, -117.2254046);

GeographicPoint end = new GeographicPoint(32.8660691, -117.217393);
```

But I get 78 versus 21 with my code, which is not of any answer. Sigh...

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DJ Djiofack Innocent · 9 days ago (/learn/advanced-data-structures/module/AfvLN/discussions/v5S9IarVEeWF2Q53QdZUbw/replies/Zj1AiAxBEeW2LwpPVHyhww)

Same here, I can't pass the question I get 78 and 20 now. And yes I am being a little bit frustrated by this

structures/profiles/a3da4d1cc426cbc43f595fde800768f0)

Cheng Chen · 9 days ago (/learn/advanced-data-structures/module/AfvLN/discussions/v5S9IarVEeWF2Q53QdZUbw/replies/Zj1AiAxBEeW2LwpPVHyhww/comments/QWPDK6xLEWhLRikesXNw)
 (/learn/advanced-data-structures/profiles/31aff96e4957af37c7f5213aea266308)
 The right answer is 82 and 19... I just randomly selected one and got it right.

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structures/profiles/f34069ce8df6de7dbefbfb7e760d9f)

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MM

Mia Minnes · Instructor · 8 days ago (/learn/advanced-data-structures/module/AfvLN/discussions/v5S9IarVEeWF2Q53QdZUbw/replies/iqdZb6zWEeWG0xjGD15hdw)
(/learn/advanced-data-structures/profiles/4c229aed8e0f158f4534a4f90386d16)
We'll look into this as soon as possible. Sorry for the frustration!

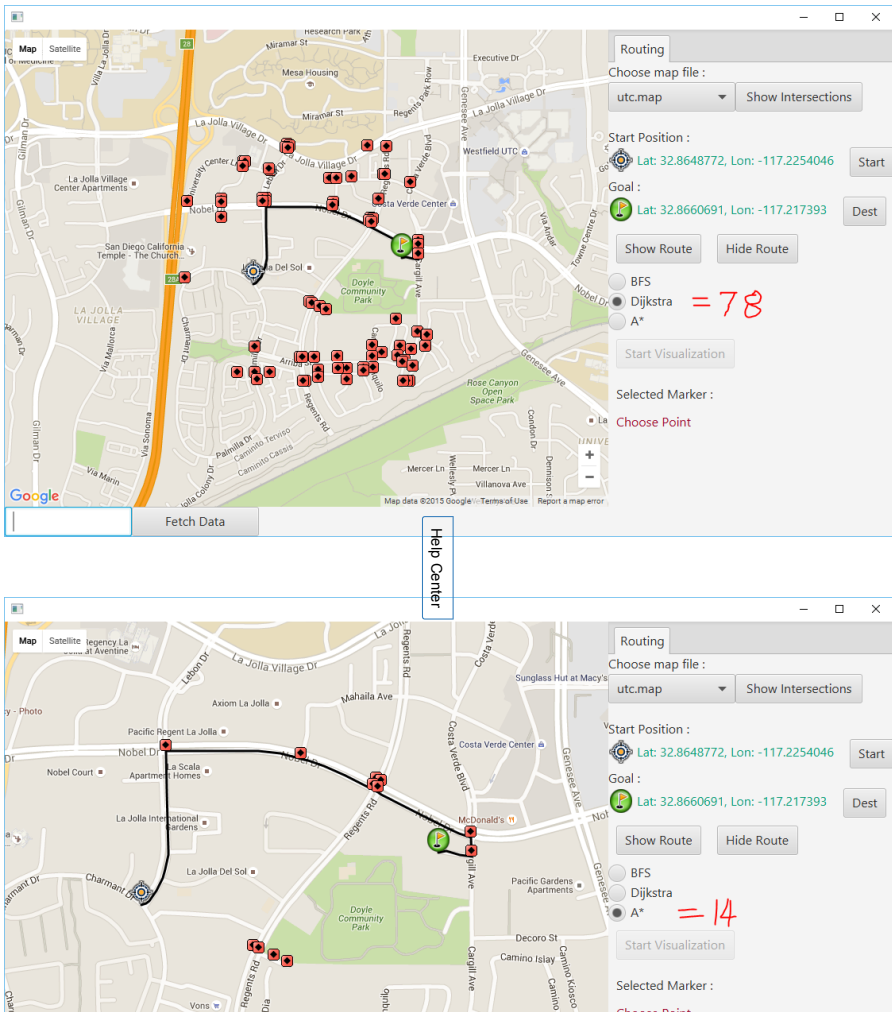
HC

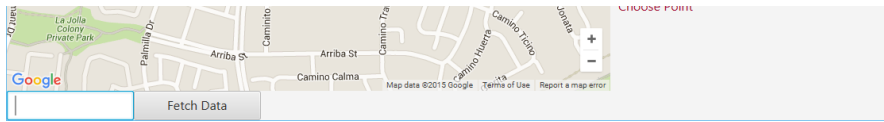
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(/learn/advanced-data-structures/profiles/f542db1ec560ecde0d6d7149d99a899)
I got 78/19, I have no idea why I'm missing 4 visited nodes for Dijkstra. Could it be the precision of my distances? I'm using "Double" to store them.

SM

Saleem Mirza · 6 days ago (/learn/advanced-data-structures/module/AfvLN/discussions/v5S9IarVEeWF2Q53QdZUbw/replies/yGh2Hq5vEWhLRikesxXNw)
(/learn/advanced-data-structures/profiles/659162581d1b3220d7c88c2d0a165395)
I'm getting 78/14 and grader is not happy. Just like Chen mentioned above, after several attempts, I selected 82/19 and got green check mark :)

I'd appreciate if teacher or someone from teaching staff can explain this behavior.





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SM

Saleem Mirza · 4 days ago (/learn/advanced-data-structures/module/AfvLN/discussions/v5S9larVEeWF2Q53QdZUbW/replies/RM_ClrAbEeWfHwqJlJBt9w)

(/learn/advanced-data-structures/profiles/659162581d1b3220d7c88c2d0a165395)
Friends, actually 82/19 is a correct answer. I'm able to match answer after little bit research.

DI

Djiofack Innocent · 4 days ago (/learn/advanced-data-structures/module/AfvLN/discussions/v5S9larVEeWF2Q53QdZUbW/replies/RM_ClrAbEeWfHwqJlJBt9w/comments/pj2lxbAvEeWacw7zR2f1aQ)

(/learn/advanced-data-structures/profiles/a3da4d1cc426cbc43f595fde800768f0)
Will be great if you can share your findings what was you doing wrong?

SZ

Sitao Zheng · 4 days ago (/learn/advanced-data-structures/module/AfvLN/discussions/v5S9larVEeWF2Q53QdZUbW/replies/RM_ClrAbEeWfHwqJlJBt9w/comments/7zckvAyEeWF6gppq4BTmQ)

(/learn/advanced-data-structures/profiles/90a227b80959bab32f9b28171e099a51)
I only get 80/19, but can pick up the right answer anyway. Maybe some minor differences with teacher's code.

SM

Saleem Mirza · 4 days ago (/learn/advanced-data-structures/module/AfvLN/discussions/v5S9larVEeWF2Q53QdZUbW/replies/RM_ClrAbEeWfHwqJlJBt9w/comments/vyKut7AzEeWixLB9mtqCw)

(/learn/advanced-data-structures/profiles/659162581d1b3220d7c88c2d0a165395)
Sure, I'm collecting diff and will post shortly.

SM

Saleem Mirza · 4 days ago (/learn/advanced-data-structures/module/AfvLN/discussions/v5S9larVEeWF2Q53QdZUbW/replies/RM_ClrAbEeWfHwqJlJBt9w/comments/S2414LA0EeWMNQ43fvjXFw) · Edited

(/learn/advanced-data-structures/profiles/659162581d1b3220d7c88c2d0a165395)
OK, wait is over. I wasn't calculating distance correctly. Here is how my new code block looks like inside loop where we search for neighbors.
Cheers!

I'm big fan of functional programming so ...

```
for (MapNode neighbor : next.getNeighbors()) {
    if (!visited.contains(neighbor)) {

        final MapNode istNode = next;
        double distanceOfNode = next.getEdges()
            .stream()
            .filter(f -> f.getEndNode() == neighbor)
            .map(m -> m.getLength() + istNode.getActualDistance())
            .min(Comparator.comparing(i -> i))
            .get();

        if (neighbor.getActualDistance() > distanceOfNode) {
            parentMap.put(neighbor, next);
            neighbor.setActualDistance(distanceOfNode);

            distanceOfNode += neighbor.getLocation().distance(endNode.getLocation());

            neighbor.setDistance(distanceOfNode);

            toExplore.offer(neighbor);
        }
    }
}
```

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SZ

Sitao Zheng · 4 days ago (/learn/advanced-data-structures/module/AfvLN/discussions/v5S9larVEeWF2Q53QdZUbW/replies/RM_ClrAbEeWfHwqJlJBt9w/comments/PbdXPrA8EeWfHwqJlJBt9w)

(/learn/advanced-data-structures/profiles/90a227b80959bab32f9b28171e099a51)
Thanks for your sharing. Beautifu stream code. One idea, this line ".min(Comparator.comparing(i -> i))" may be replaced by ".min(Comparator.naturalOrder())" to have a more clear meaning?

And I don't use a MapNode class so it seems there are some differences between our code in this part, the basic idea is same however:)

▲ 0 Upvote

SM

Saleem Mirza · 3 days ago (/learn/advanced-data-structures/module/AfvLN/discussions/v5S9larVEeWF2Q53QdZUbW/replies/RM_ClrAbEeWfHwqJlJBt9w/comments/t9Klg7B9EeWjcBKJYq1ZMQ)

(/learn/advanced-data-structures/profiles/659162581d1b3220d7c88c2d0a165395)
Indeed, in current context it can be replaced. Thank you!

SL

Sylvain Leroux · 2 days ago (/learn/advanced-data-structures/module/AfvLN/discussions/v5S9larVEeWF2Q53QdZUbW/replies/RM_ClrAbEeWfHwqJlJBt9w/comments/Bh8g07FjEeWMNQ43fvjXFw) · Edited

(/learn/advanced-data-structures/profiles/659162581d1b3220d7c88c2d0a165395)
@Saleem
I'm not sure about the "f.getEndNode().equals(neighbor)" part.

As those are objects, you shouldn't compare for *identity* but for *equality*: "f.getEndNode().equals(neighbor)"

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SD Reply
(/learn/advanced-data-structures/profiles/f34069ce8df6de7dbebfedfb7e760d9f)

Dijofack Innocent · 2 days ago (/learn/advanced-data-structures/module/AfvLN/discussions/v5S9IarVeeWF2Q53QdUbw/replies/UPsGhrEmEWhLRikesxXNW/comments/aYI1ObE0EwRTArgYkByQ)

/learn/advanced-data-structures/module/AfvLN/discussions/v5S9IarVeeWF2Q53QdUbw/replies/UPsGhrEmEWhLRikesxXNW/comments/aYI1ObE0EwRTArgYkByQ)

I am still trying to figure it out. will look at it again shortly

es/profiles/a3da4d1cc426cbc43f595fde800768f0)

RP Rohan Paul · 2 days ago (/learn/advanced-data-structures/module/AfvLN/discussions/v5S9IarVeeWF2Q53QdZUbw/replies/UPsGhrEmEeWhLRikesxXNw/comments/hoK1cLFJEeWF6gpp4BTmQ) Hi guys!!

/learn/advanced-data-structures/profiles/45edma432433e684f6e6b4cfe12045b that the closest option was 82/19 I marked it.

When comparing the new distance with the old distance I was doing a check of the following type :

Now, I changed this to :

```
if(newDistance <= oldDistance)
```

and voila!! I got 82/19!!

So, basically they want us to update the entries in the priority queue even if the distance to the next node via the current node is equal to the distance for the next node found so far.

Hope this helps!!

PS : I am not sure if I should have posted this here; If it's against any policy, feel free to point that out and I'll remove it.

DI [Dijofack Innocent · 2 days ago \(/learn/advanced-data-structures/module/AfvLN/discussions/v5S9IarVeeWF2Q3ZQsdZUbw/replies/UPsGhrEmEeWhLRikesxNw/comments/zaGsJLFREeWMNQ43fvjXfw\)](#)

(/learn/advanced-data-structures/profile/a3da4d1cc426cbc43f595fde800768f0)

Ray Stinson · 2 days ago (/learn/advanced-data-structures/module/AFvLN/discussions/v559iarVEeWF2Q53QdZUbw/replies/UPsGhrEmEeWhLRikesxXNw/comments/u800l7FsEeWoG6ulZMPew) · Edited

/learn/advanced-data-structures/profiles/28730dc0139ea780450918eca787dae0)

Thanks, Kohan! I now have similar results after allowing equality.

0 Upvote

 Zhen Yan · 2 days ago (/learn/advanced-data-structures/module/AfvLN/discussions/v5S9larVeeWf2Q53QdZUbw/replies/UPsGhrEmEeWlhRikesxNw/comments/xoMRf7GGEeWf6gqp48TmQJ)
(/learn/advanced-data-structures/profiles/364ffa2e1ea55f7a9309f9b13983af6f)
I think you are right. Thank you, Rohan!